15/9/23

Mathematics

Section-A

1.

3	36 - 45
3	12 - 15
2	4 - 5
2	2 - 5
5	1 - 5
	1 - 1

= 180 sec.

(3) = 3 minutes.

2. $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ} = \frac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\frac{2}{\sqrt{3}}}{1-\frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}}$

$= \frac{\cancel{2} \times \cancel{3}}{\cancel{\sqrt{3}} \times \cancel{2}} = \sqrt{3}$

(3) $\tan 60^\circ$.

3. $\frac{AP}{PB} = \frac{PD}{PC}$.

$\angle APB = \angle DPC$ (VOA)

$\Delta APB \sim \Delta DPC$. (SAS ~)

$\angle PBA = 180^\circ - (50^\circ + 30^\circ) = 180^\circ - 100^\circ$

(4) $\angle PBA = 100^\circ$

4.

4. $\frac{ar\,\Delta ABC}{ar\,\Delta QRP} = \frac{9}{4} = \frac{BC^2}{PR^2}$

$\frac{9}{4} = \frac{15^2}{PR^2}$

$PR^2 = \frac{4 \times \cancel{15}^{5} \times \cancel{15}^{5}}{\cancel{9}\,\cancel{3}} = 100$

(1) $PR = 10\,cm.$

5. $(2)^2 + (a+1)2 + b = 0$

$4 + 2a + 2 + b = 0.$

$2a + b + 6 = 0$

$2a + b = -6$ — ①

$(-3)^2 + (a+1)(-3) + b = 0$

$9 - 3a - 3 + b = 0$

$-3a + b + 6 = 0$

$-3a + b = -6$ — ②

$2a + \cancel{b} = -6$

$-3a + \cancel{b} = -6$

$+ \quad - \quad +$

$-a = 0$

(4) $\boxed{a = 0.}$

2	1250
5	625
5	125
5	25
5	5
	1

6. $\frac{14587}{1250} = \frac{14587}{2 \times 5^4}$

$= \frac{14587}{10 \times 5^3} = \frac{14587}{10 \times 1000}$

(4) = four decimal places

7.

7. $a - b = 2$ — ①

$a + b = 4$ — ②

② − ①

$a + b = 4$

$a - b = 2$

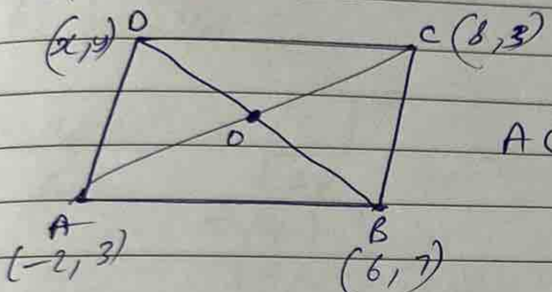
$\underline{- \quad + \quad -}$

$2b = 2$

$b = 1$

$a = 3.$

(3) 3 and (1)

8. 

AC and BD bisect at O.

mid point AC = mid point BD.

$$\left[\left(\frac{-2+8}{2}\right), \left(\frac{3+3}{2}\right)\right] = \left[\left(\frac{6+x}{2}\right), \left(\frac{7+y}{2}\right)\right]$$

$$\frac{6}{2}, \frac{6}{2} = \frac{6+x}{2}, \frac{7+y}{2}$$

$$6 + x = 6$$

$$\boxed{x = 0}$$

$$7 + y = 6$$

$$y = -1$$

= (2) D(0, −1)

9. $\frac{a_1}{a_2} = \frac{\cancel{9}\,1}{\cancel{18}\,2}, \frac{b_1}{b_2} = \frac{\cancel{3}\,1}{\cancel{6}\,2}, \frac{c_1}{c_2} = \frac{\cancel{12}\,6}{\cancel{26}\,13}$

$\frac{1}{2} = \frac{1}{2} \neq \frac{6}{13}$

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(d) No solution

10. $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\frac{2}{3} = \frac{-3}{k}$

$2k = -9$

(A) $k = \frac{-9}{2}$

$= (1)\ \frac{-9}{2}$

11. $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

$\frac{10}{6} \neq \frac{-5}{k}$

$10k \neq -30$

(1) $k \neq -3.$

12. $616 = 2 \times 2 \times 2 \times 7 \times 11$

$32 = 2 \times 2 \times 2 \times 2 \times 2$

$= 8.$

(1) 8

2	616
2	308
2	154
7	77
11	11
	1

13. (1) similar

14. (2) x^2yz^2

15. ~~x~~ sum $= 4-3=1$

product $= -12$

$x^2-5x+4 = x^2-x-12$

(3) $\frac{x^2}{2}-\frac{x}{2}-6 \Rightarrow x^2-x-12$

16. (3) intersecting or coincident

17. (1) parallel

18. $\frac{ar\,\Delta_1}{ar\,\Delta_2} = \left(\frac{S_1}{S_2}\right)^2$

$= \left(\frac{4}{9}\right)^2 = \frac{16}{81}$

(4) 16 : 81

19. x^2-4x-5 , $25-20-5=0$

$1+\cancel{4}-5=0$

$2+\cancel{\sqrt{3}}+2-\cancel{\sqrt{3}} = 4$ x^2-4x+1

$4-3 = 1$

(3) A is true but R is false.

20. $\frac{3(k-1)}{15} = \frac{4}{20\,5}$ $\cancel{5}(3k-3) = \cancel{15}\,3$

$3k=6$

$k=2$

(1) Both A and R are true and R is the correct explanation of A.

Section-B

21. $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$= 5(6 \times 4 \times 3 \times 2 \times 1 + 1)$

$= 5(144 + 1)$

$= 5(145)$. $= 5(144) + 5(1)$

$24 \times 6 = 144$

$145 \times 5 = 725$

$144 \times 5 = 720$

as $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$.

has more than two factor i.e. ~~5, 145~~ 5, 144, 1

so it a composite no.

22. $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$4x + 6y = 7$ — (1)

comparing with $a_1x + b_1y + c_1 = 0$.

$a_1 = 4$, $b_1 = 6$, $c_1 = -7$

$kx + 18y = 21$ — (2)

comparing with $a_2x + b_2y + c_2 = 0$

$a_2 = k$, $b_2 = 18$, $c_2 = -21$

$\frac{4}{k} = \frac{\cancel{6}^{1}}{\cancel{18}_{3}} = \frac{7}{21}$

$\boxed{k = 12}$

23. $(0,0)$ and $(-6,8)$

using distance formula $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{36 + 64} = \sqrt{100}$

$= 10$ units

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24. $\sin 3A = \cos(A - 10^\circ)$

$\cos(90^\circ - 3A) = \cos(A - 10^\circ)$

$90^\circ - 3A = A - 10^\circ$
$90^\circ + 10^\circ = A + 3A$
$4A = 100^\circ$
$\boxed{A = 25^\circ}$

25. AB ∥ CD (given)
In △ AOB and △DOC,

∠AOB = ∠DOC (VOA)
∠OAB = ∠ODC (alt. angles)

from (AA ∼), △AOB ∼ △DOC

26.

Section - C

26. $\dfrac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \dfrac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)} = 2\operatorname{cosec}\theta$

$\underline{LHS} = \dfrac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \dfrac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)}$

$= \dfrac{\sin\theta}{1 + \cos\theta} + \dfrac{1 + \cos\theta}{\sin\theta}$

$= \dfrac{\sin^2\theta + (1 + \cos\theta)^2}{(1 + \cos\theta)(\sin\theta)} = \dfrac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{(1 + \cos\theta)\sin\theta}$

$= \frac{1+1+2\cos\theta}{(1+\cos\theta)\sin\theta}$ $(\sin^2\theta + \cos^2\theta = 1)$

$= \frac{2+2\cos\theta}{(1+\cos\theta)\sin\theta}$

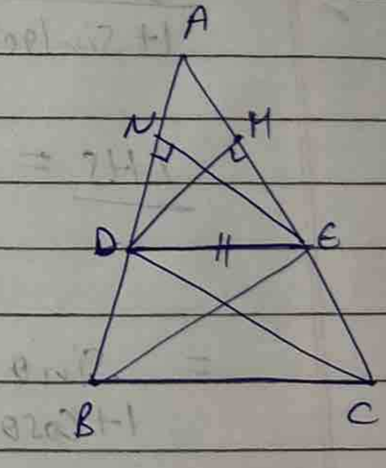
$= \frac{2\cancel{(1+\cos\theta)}}{\cancel{(1+\cos\theta)}\sin\theta}$

$= \frac{2}{\sin\theta}$ $\left(\frac{1}{\sin\theta} = \operatorname{cosec}\theta\right)$

$= 2\operatorname{cosec}\theta = RHS$

$LHS = RHS$

27. According to basic Proportionality Theorem if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides of the triangle in proportion.



Given :- $DE \parallel BC$

To prove :- ~~DB~~ $\frac{AD}{DB} = \frac{AE}{EC}$

Construction :- Draw $DM \perp AC$
$EN \perp AB$

Proof :- Join to D to C and B to E

$ar\ \Delta ADE = \frac{1}{2} \times AD \times DM$

$$ar\ \Delta DBE = \frac{1}{2} \times DB \times DH$$

$$\frac{ar\ of\ \Delta ADE}{ar\ of\ \Delta DBE} = \frac{\cancel{\frac{1}{2}} \times AD \times \cancel{DH}}{\cancel{\frac{1}{2}} \times DB \times \cancel{DH}} \quad \text{--- (1)}$$

$$\frac{ar\ \Delta ADE}{ar\ \Delta DBE} = \frac{AD}{DB} \quad \text{--- (2)}$$

Similarly, $\frac{ar\ \Delta ADE}{ar\ \Delta DCE} = \frac{\cancel{\frac{1}{2} \times EN \times DE}}{\cancel{\frac{1}{2} \times EN \times D}}$ $\frac{\cancel{\frac{1}{2}} \times AE \times \cancel{DH}}{\cancel{\frac{1}{2}} \times EC \times \cancel{DH}}$ --- (3)

$$= \frac{AE}{EC} \quad \text{--- (4)}$$

ar ΔDEB and ΔDEC is same because Δ's lie on same base DE and between same ||s.

$$\therefore ar\ DEB = ar\ \Delta DEC \quad \text{--- (5)}$$

from (2), (4) and (5).

$$\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$$

Hence proved.

28. let fixed charge for first two days $= x$.
additional charge for each day $= y$.

Acc. to que, $x + 4y = 22$ --- (1)

Anand, $x + 2y = 16$ — ②

① − ②

$$\begin{array}{r} x + 4y = 22 \\ -x + 2y = 16 \\ \hline 2y = 6 \end{array}$$

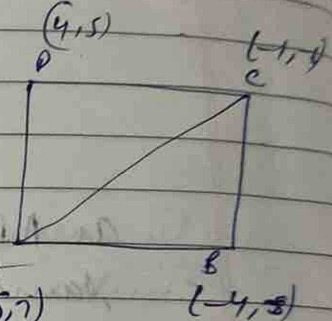
$y = 3$

$x = 10$

∴ fixed charge is ₹10 while additional charge is ₹3.

30

29. $A(-5,7)$ $B(-4,-5)$ $C(-1,-6)$ $D(4,5)$.



area of ABCD = area of ΔABC + area of ΔADC

area of $\Delta ABC = \frac{1}{2}\left(x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right)$

$x_1 = -5,\ y_1 = 7,\ x_2 = -4,\ y_2 = -5,\ x_3 = -1,\ y_3 - 6$

area $\Delta ABC = \frac{1}{2}\left[-5(-5+6) - 4(-6-7) - 1(7+5)\right]$

$= \frac{1}{2}(35) = \frac{35}{2}$ sq. units

area of $\Delta ADC = \frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$

$x_1 = -5,\ y_1 = 7,\ x_2 = 4,\ y_2 = 5,\ x_3 = -1,\ y_3 = 6.$

area of $\Delta ADC = \frac{1}{2}\left[-5(5+6) + 4(-6-7) - 1(7-5)\right] = \frac{1}{2}(-109) = \frac{109}{2}$ sq. units

$\therefore$ area of ABCD $= \frac{35}{2} + \frac{109}{2} = \frac{144}{2}$

$= 72$ sq units

30. Using Euclid's algorithm :-

$1353 = 861 \times 1 + 492$
$861 = 492 \times 1 + 369$
$492 = 369 \times 1 + 123$
$369 = 123 \times 3 + 0.$

So HCF, 1353, 861 = 123

31. point = M $(x, 0)$.

A $(5, -2)$, B$(-3, 2)$

using distance formula,
AM = BM.

$$\sqrt{(x-5)^2 + (-2)^2} = \sqrt{(x+3)^2 + (2)^2}$$

$$(x-5)^2 + (-2)^2 = (x+3)^2 + (2)^2$$
$$\cancel{x^2} + 25 - 10x + 4 = \cancel{x^2} + 9 + 6x + 4$$
$$-10x + 29 = 13 + 6x$$
$$-10x - 6x = 13 - 29$$
$$\cancel{-}16x = \cancel{-}16$$
$$\boxed{x = 1}$$

point is $(1, 0)$.

Section - D

32. $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1+\sec\theta\ \text{cosec}\,\theta$

LHS :- $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$

$$= \frac{\frac{\sin\theta}{\cos\theta}}{1-\frac{\cos\theta}{\sin\theta}} + \frac{\frac{\cos\theta}{\sin\theta}}{1-\frac{\sin\theta}{\cos\theta}}$$

$$= \frac{\frac{\sin\theta}{\cos\theta}}{\frac{\sin\theta-\cos\theta}{\sin\theta}} + \frac{\frac{\cos\theta}{\sin\theta}}{\frac{\cos\theta-\sin\theta}{\cos\theta}}$$

$$= \frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} + \frac{\cos^2\theta}{\cos\theta(\cos\theta-\sin\theta)}$$

$$= \frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \frac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)}$$

$$= \frac{\sin^3\theta-\cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$$

$$= \frac{\cancel{(\sin\theta-\cos\theta)}(\sin^2\theta+\cos^2\theta+\sin\theta\cdot\cos\theta)}{\cancel{(\sin\theta-\cos\theta)}(\sin\theta\cdot\cos\theta)}$$

$$= \frac{\sin^2\theta+\cos^2\theta+\sin\theta\cos\theta}{\sin\theta\cdot\cos\theta}.$$

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$$= \frac{\sin^2\theta}{\sin\theta \cdot \cos\theta} + \frac{\sin\theta \cdot \cos\theta}{\sin\theta \cdot \cos\theta} + \frac{\cos^2\theta}{\sin\theta \cdot \cos\theta}$$

$$= \frac{\sin\theta}{\cos\theta} + 1 + \frac{\cos\theta}{\sin\theta}$$

$$\cancel{= \tan\theta + 1 + \cot\theta.}$$

$$= 1 + \frac{\sin^2\theta + \cos^2\theta}{\sin\theta \cdot \cos\theta}$$

$$= 1 + \frac{1}{\cos\theta \cdot \sin\theta}$$

$$= 1 + \sec\theta \cdot \operatorname{cosec}\theta.$$

$\therefore$ LHS = RHS

Hence proved.

33. $x - y = -1$ —— (1)

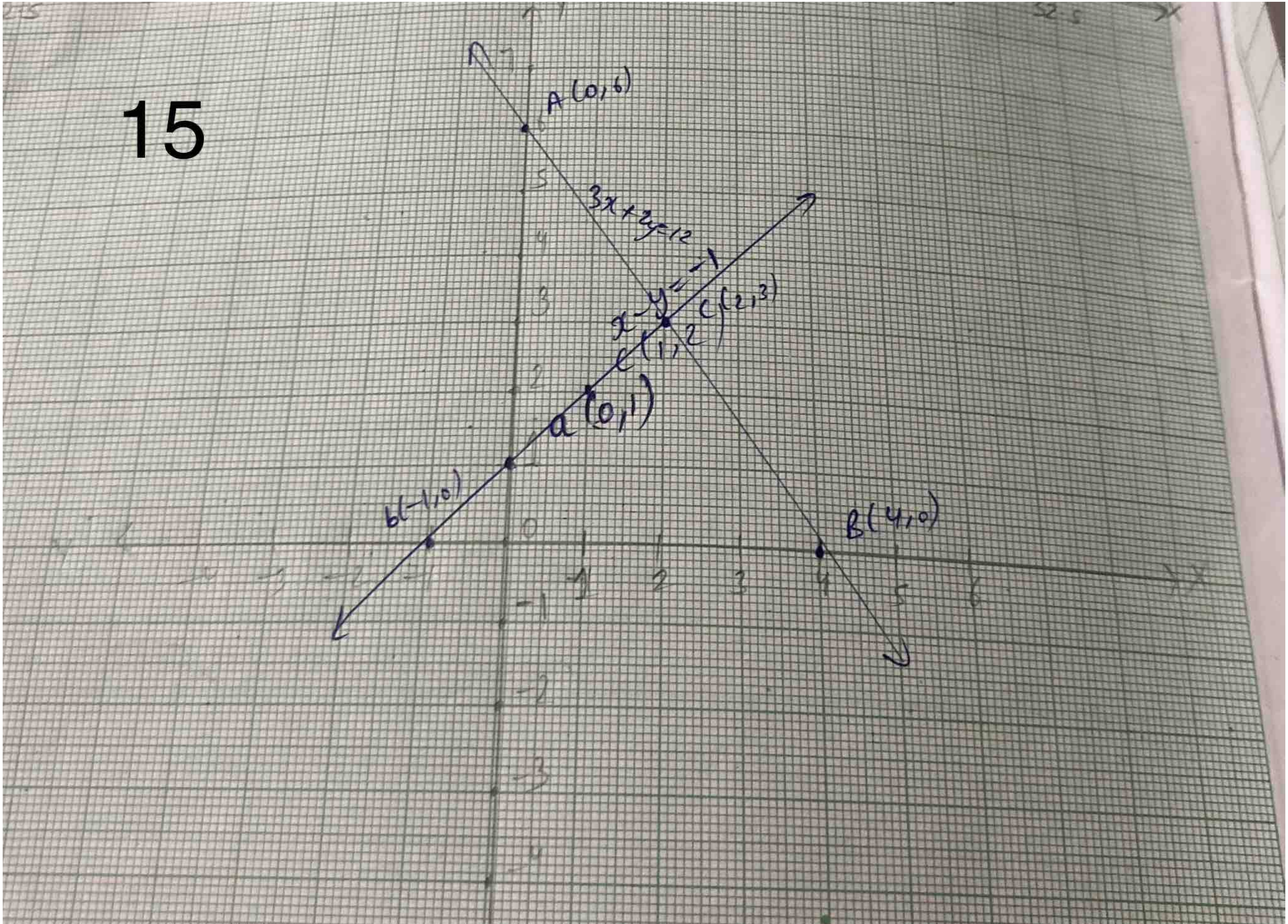
x	0	-1	1
y	1	0	2

$3x + 2y = 12$ —— (2)

33.

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 5 \times 3 = \frac{15}{2} \text{ sq. units}$$


A(0,6)
3x+2y=12
x−y=−1
C(2,3)
c(1,2)
a(0,1)
b(−1,0)
B(4,0)
y
x

34. $2x^3 - x^2 - 4x + 2$

let α, β, γ are the zeroes of polynomi
and $\sqrt{2}, -\sqrt{2}$ are zeros.

$$\alpha + \beta + \gamma = \frac{-b}{a} = \frac{-(-1)}{2} = \frac{1}{2}$$

$$\sqrt{2} + (-\sqrt{2}) + \gamma = \frac{1}{2}$$

$$\gamma = \frac{1}{2}.$$

So other zero is $\frac{1}{2}$.

35. $A(4,-3)$ $B(8,5)$

$m_1 : m_2 = 3:1$

let point be (x,y)

Using section formula,

$x = \frac{m_1x_2 + m_2x_1}{m_1+m_2}$; $y = \frac{m_1y_2 + m_2y_1}{m+m_2}$

$x = \frac{3(8)+1(4)}{4}$; $y = \frac{3(5)+1(-3)}{4}$

$x = \frac{24+4}{4} = \frac{28}{4}$; $y = \frac{15-3}{4} = \frac{12}{4}$

$x = 7$; $y = 3$

point is $(7,3)$.

Section-E

36. (i) ~~(1)(12,0)~~ (2) $(0,12)$

(ii) (1) $(3,5)$

(iii) $P(3,5)$ and $Q(6,9)$

$\sqrt{(6-3)^2+(9-5)^2} = \sqrt{9+16} = \sqrt{25} = 5$ units

(3) 5m

~~(iv)~~ 37. (i) (3) Parabola

(ii) x^2+x-6.
$\cancel{x^2-2x-3}$, $\cancel{9+6-3}$ = $9-3-6=0$. (3) 0

(iii) (3) $-3, 2$.

38. (i) Kite is a rhombus.
$\frac{1}{\cancel{2}} \times \cancel{6}^{3} \times 8 = 24 cm^2$
(3) $24 cm^2$

(ii) (1) RHS

(iii) $\frac{ar(\Delta ABC)}{ar(\Delta DEF)}$ = Square of ratio of ~~two~~ two corresponding medians

(2) 4:9